

# WHAT TO EXPECT FOR (AND HOW TO PREPARE FOR) THE FINAL EXAM

P8107

Fall 2025

# Format

- The exam will consist of several multi-part "free response"-type questions.
- Next Monday (Dec. 8) 9:00am-12:00noon
- Each student may use up to **four** pages of notes (front and back – A4 or 8.5 × 11) during the exam. No other resources\* are permitted.
- Emphasis is on understanding and applying concepts (not on calculation).

\*An "old school" calculator is OK – but nothing that connects to the internet

# What material will be covered?

- Everything covered in class
- Emphasis is on “new material”:
  - Chapter 8 Estimation (P9.1 – P9.5)
  - Chapter 9 Properties of point estimators (P10.1 – P10.5)
  - Chapter 10 Hypothesis testing (P11.1 – P11.6)
- You will not be expected to know anything in the book that has not been covered in the videos.

# Topics

- Point estimation
  - Bias, MSE, (relative) efficiency, consistency, asymptotically unbiased
- Confidence intervals
  - Pivotal quantity, large-sample CI's, small-sample CI's, CI's for variance
- Maximum likelihood estimation
  - Methods of finding, invariance, asymptotic distribution
- Hypothesis testing
  - Type I, Type II errors, power, p-values, Neyman-Pearson Lemma, LR tests, asymptotic distribution of LR test statistics

# How to prepare?

- Review lecture examples, in-class problems, and homework problems
- Work sample final problems BUT THIS WILL NOT BE SUFFICIENT!
- AND **DO NOT EXPECT** THE PROBLEMS ON THE EXAM TO BE “JUST LIKE” THE SAMPLE PROBLEMS!
- Work extra problems!
- Working the assigned problems is necessary but not sufficient to master the material.
- Try “Supplementary exercises” at the end of each chapter.

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$$Y_1, \dots, Y_n \sim \text{iid } N(\mu, \sigma^2)$$

$$H_0: \mu = \mu_0$$

$$H_a: \mu \neq \mu_0$$

Find LR test.

Hint:  $\bar{Y}$  and  $s^2$  are independent

$$\begin{aligned} L(\mu, \sigma^2) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - \mu)^2}{2\sigma^2}} \\ &= (2\pi)^{-\frac{n}{2}} (\sigma^2)^{-\frac{n}{2}} e^{-\frac{1}{2\sigma^2} \sum (y_i - \mu)^2} \end{aligned}$$

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2$$

$$- \frac{1}{2\sigma^2} \sum (y_i - \mu)^2$$

$$\frac{\partial}{\partial \mu} \ell(\mu, \sigma^2) = -\frac{1}{2\sigma^2} \sum 2(y_i - \mu)(-1) = \frac{1}{\sigma^2} \sum (y_i - \mu)$$



$$\frac{\partial}{\partial \mu} l(\mu, \sigma^2) \stackrel{\text{set}}{=} 0$$

$$\sum (y_i - \mu) = 0$$

$$\sum y_i - n\mu = 0$$

$$n\mu = \sum y_i$$

$$\mu = \frac{1}{n} \sum y_i = \bar{y}$$

unrestricted  
MLE for  
 $\mu$

$$\begin{aligned} \frac{\partial}{\partial \sigma^2} l(\mu, \sigma^2) &= \frac{\partial}{\partial \sigma^2} \left( -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 \right. \\ &\quad \left. - \frac{1}{2\sigma^2} \sum (y_i - \mu)^2 \right) \end{aligned}$$

$$= -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum (y_i - \mu)^2 \stackrel{\text{set}}{=} 0$$

$$-\frac{n}{2} + \frac{1}{2\sigma^2} \sum (y_i - \mu)^2 = 0$$

$$\frac{1}{2\sigma^2} \sum (y_i - \mu)^2 = \frac{n}{2}$$

$$\sigma^2 = \frac{1}{n} \sum (y_i - \mu)^2$$

So  $\hat{\sigma}^2 = \frac{1}{n} \sum (y_i - \hat{\mu})^2 = \frac{1}{n} \sum (y_i - \bar{y})^2$   
 is the unrestricted MLE for

Restricted MLE is  $\hat{\mu} = \mu_0$

$$\hat{\sigma}_R^2 = \frac{1}{n} \sum (y_i - \mu_0)^2$$

$$\frac{L(\Omega_0)}{L(\Omega)} = \frac{(2\pi)^{-\frac{n}{2}} (\hat{\sigma}_R^2)^{-\frac{n}{2}} e^{-\frac{1}{2\hat{\sigma}_R^2} \sum (y_i - \mu_0)^2}}{(2\pi)^{-\frac{n}{2}} (\hat{\sigma}^2)^{-\frac{n}{2}} e^{-\frac{1}{2\hat{\sigma}^2} \sum (y_i - \bar{y})^2}}$$

$$-\frac{1}{2\hat{\sigma}^2} \sum (y_i - \mu_0)^2 = -\frac{n}{2}$$

$$\frac{(\hat{\sigma}_R^2)^{-\frac{n}{2}}}{(\hat{\sigma}^2)^{-\frac{n}{2}}} \frac{e^{-\frac{n}{2}}}{e^{-\frac{n}{2}}}$$

Reject  $H_0$  if  $\left( \frac{\hat{\sigma}_R^2}{\hat{\sigma}^2} \right)^{-\frac{n}{2}} < K$

$$\text{if } \frac{\hat{\sigma}_R^2}{\hat{\sigma}^2} < k^{-\frac{2}{n}}$$

$$\text{if } \frac{\sum (y_i - \mu_0)^2}{\sum (y_i - \bar{y})^2} < k^{-\frac{2}{n}}$$

Let's eat more donuts...

$$\text{if } \frac{\sum (y_i^2 - 2\mu_0 y_i + \mu_0^2)}{n \hat{\sigma}^2} < k^{-\frac{2}{n}}$$

$$\text{if } \frac{\sum y_i^2 - 2n\mu_0 \bar{y} + n\mu_0^2}{n \hat{\sigma}^2} < k^{-\frac{2}{n}}$$

$$\text{if } n \hat{\sigma}^2 k^{-\frac{2}{n}} > \sum y_i^2 - 2n\mu_0 \bar{y} + n\mu_0^2$$

Problem:  $Y_1, Y_2, \dots, Y_n \sim \text{iid } N(\mu, \sigma^2)$ . Find a LR test for  $H_0: \mu = \mu_0$   
 vs.  $H_a: \mu \neq \mu_0$

HINT:  $\bar{Y}$  and  $s^2$   
 are independent

2 free parameters

Unrestricted parameter space is  $\Omega = \{(\mu, \sigma^2), -\infty < \mu < \infty, \sigma^2 > 0\}$

and restricted parameter space is  $\Omega_0 = \{(\mu, \sigma^2), \mu = \mu_0, \sigma^2 > 0\}$   
 1 free parameter

Unrestricted:

[I'm going to use  $\nu$  in place of  $\sigma^2$  - it might seem strange to take derivatives w.r.t.  $\sigma^2$ ]

$$L(\mu, \nu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\nu}} \frac{1}{\sqrt{\nu}} e^{-\frac{(y_i - \mu)^2}{2\nu}} = (2\pi)^{-\frac{n}{2}} \nu^{-\frac{n}{2}} e^{-\frac{\sum (y_i - \mu)^2}{2\nu}}$$

$$\ell(\mu, \nu) = \log L(\mu, \nu) = -\frac{n}{2} \log 2\pi - \frac{n}{2} \log \nu - \frac{1}{2\nu} \sum (y_i - \mu)^2$$

$$\frac{\partial}{\partial \mu} \ell(\mu, \nu) = -\frac{1}{2\nu} \sum \frac{\partial}{\partial \mu} (y_i - \mu)^2 = -\frac{1}{2\nu} \sum 2(y_i - \mu)(-1) = \frac{1}{\nu} \sum (y_i - \mu) \stackrel{\text{set}}{=} 0$$

$$\sum y_i - n\mu = 0$$

$$\hat{\mu} = \frac{\sum y_i}{n} = \bar{y}$$

(no surprise here)

$$\frac{\partial}{\partial \nu} \ell(\mu, \nu) = -\frac{n}{2\nu} + \frac{1}{2\nu^2} \sum (y_i - \mu)^2 \stackrel{\text{set}}{=} 0$$

(technically, should check 2nd deriv....)

$$\frac{n}{2\nu} = \frac{1}{2\nu^2} \sum (y_i - \mu)^2$$

$$\nu = \frac{1}{n} \sum (y_i - \mu)^2 \rightarrow \text{plug in MLE for } \mu$$

$$\hat{\nu} = \frac{1}{n} \sum (y_i - \bar{y})^2$$

To get the MLE for the restricted case,  $\hat{\mu} = \mu_0$  (only option!)

and so  $\hat{\nu}_0 = \frac{1}{n} \sum (y_i - \mu_0)^2$  (Same calculations as above)

the subscript 0 indicates  $\Omega_0$

OK, now we plug these MLEs into the likelihood function:

$$L(\hat{\Omega}_0) = (2\pi)^{-\frac{n}{2}} \hat{\nu}_0^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\hat{\nu}_0} \sum (y_i - \mu_0)^2\right\}$$

$$= (2\pi)^{-\frac{n}{2}} \hat{\nu}_0^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\hat{\nu}_0} \sum (y_i - \mu_0)^2\right\} = (2\pi)^{-\frac{n}{2}} \hat{\nu}_0^{-\frac{n}{2}} e^{-\frac{n}{2}}$$

Then the unrestricted case:

$$\begin{aligned}
 L(\hat{\Sigma}) &= (2\pi)^{-\frac{n}{2}} \hat{\nu}^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\hat{\nu}} \sum (y_i - \bar{y})^2\right\} \\
 &= (2\pi)^{-\frac{n}{2}} \hat{\nu}^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\hat{\nu} \cancel{\sum (y_i - \bar{y})^2}} \cancel{\sum (y_i - \bar{y})^2}\right\} \\
 &= (2\pi)^{-\frac{n}{2}} \hat{\nu}^{-\frac{n}{2}} e^{-\frac{n}{2}}
 \end{aligned}$$

(Kinda neat how these terms always cancel, huh?)

Likelihood ratio:

$$\begin{aligned}
 \lambda &= \frac{L(\hat{\Sigma}_0)}{L(\hat{\Sigma})} = \frac{(2\pi)^{-\frac{n}{2}} \hat{\nu}_0^{-\frac{n}{2}} e^{-\frac{n}{2}}}{(2\pi)^{-\frac{n}{2}} \hat{\nu}^{-\frac{n}{2}} e^{-\frac{n}{2}}} = \left(\frac{\hat{\nu}_0}{\hat{\nu}}\right)^{-\frac{n}{2}} \\
 &= \left(\frac{\sum (y_i - \mu_0)^2}{\sum (y_i - \bar{y})^2}\right)^{-\frac{n}{2}}
 \end{aligned}$$

Reject  $H_0$  if this is "small"

Here's the trick:  $\sum (y_i - \mu_0)^2 = \sum (y_i - \bar{y} + \bar{y} - \mu_0)^2$

$$\begin{aligned}
 &= \sum (y_i - \bar{y})^2 + \sum (\bar{y} - \mu_0)^2 + 2\sum (\bar{y} - \mu_0)(y_i - \bar{y}) \\
 &= \sum (y_i - \bar{y})^2 + n(\bar{y} - \mu_0)^2 + 2(\bar{y} - \mu_0)\sum (y_i - \bar{y}) \\
 &= \sum (y_i - \bar{y})^2 + n(\bar{y} - \mu_0)^2 + 2(\bar{y} - \mu_0)\underbrace{\left(\sum y_i - n\bar{y}\right)}_{\substack{\text{since } \sum y_i = n\bar{y} \\ \text{this thing is zero}}}
 \end{aligned}$$

$$\lambda = \left(\frac{\sum (y_i - \bar{y})^2 + n(\bar{y} - \mu_0)^2}{\sum (y_i - \bar{y})^2}\right)^{-\frac{n}{2}} = \left(1 + \frac{n(\bar{y} - \mu_0)^2}{\sum (y_i - \bar{y})^2}\right)^{-\frac{n}{2}}$$

Reject  $H_0$  if  $\lambda < k$

if  $\left(1 + \frac{n(\bar{y} - \mu_0)^2}{\sum (y_i - \bar{y})^2}\right)^{-\frac{n}{2}} < k$

if  $1 + \frac{n(\bar{y} - \mu_0)^2}{\sum (y_i - \bar{y})^2} > k^{-\frac{2}{n}}$

the inequality switches because we raised both sides to a negative power

if  $\frac{(\bar{y} - \mu_0)^2}{\frac{1}{n} \sum (y_i - \bar{y})^2} > k^{-\frac{2}{n}} - 1$

e.g.  $2 < 3$   
 $2^{-1} > 3^{-1}$      $\frac{1}{2} > \frac{1}{3}$

Or, reject  $H_0$  if  $\frac{\bar{y} - \mu_0}{\sqrt{\frac{1}{n} \sum (y_i - \bar{y})^2}} < -\sqrt{k^{-\frac{2}{n}} - 1}$  or if  $\frac{\bar{y} - \mu_0}{\sqrt{\frac{1}{n} \sum (y_i - \bar{y})^2}} > \sqrt{k^{-\frac{2}{n}} - 1}$

Can we figure out the distribution of this thing?

If  $H_0$  is true,  $\bar{Y} - \mu_0 \sim N(0, \frac{\sigma^2}{n})$

$$\frac{\bar{Y} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$$

Also we know that  $\frac{\sum (Y_i - \bar{Y})^2}{\sigma^2} \sim \chi_{n-1}^2$

Reject  $H_0$  if  $\frac{\bar{Y} - \mu_0}{\sqrt{\frac{1}{n} \sum (y_i - \bar{y})^2}} < -\sqrt{k^{-\frac{2}{n}} - 1}$

if  $\frac{\frac{\bar{Y} - \mu_0}{\sigma/\sqrt{n}}}{\sqrt{\frac{\frac{1}{n} \sum (y_i - \bar{y})^2}{\sigma^2}}} < -\sqrt{k^{-\frac{2}{n}} - 1}$

if  $\frac{\frac{\bar{Y} - \mu_0}{\sigma/\sqrt{n}}}{\sqrt{\frac{\sum (Y_i - \bar{Y})^2}{\sigma^2} / (n-1)}} < -\sqrt{(n-1)(k^{-\frac{2}{n}} - 1)}$

$N(0,1)$   $\nearrow$   
 $\sqrt{\chi_{df}^2/df}$   $\nearrow$   
 so this has a  $t_{df}$  distribution!  
 (if  $H_0$  is true!)

So if  $t =$  that thing, we need to choose the cut-off  
 so that  $P(\text{Type I error}) = \alpha$

$$P(t < -t_{\frac{\alpha}{2}}) + P(t > t_{\frac{\alpha}{2}}) = \frac{\alpha}{2} + \frac{\alpha}{2} = \alpha$$

Now, we can simplify back -

$$t = \frac{\bar{Y} - \mu_0}{\frac{\sigma/\sqrt{n}}{\sqrt{\frac{\sum (Y_i - \bar{Y})^2}{\sigma^2} / (n-1)}}} = \frac{\sqrt{n} (\bar{Y} - \mu_0)}{\sqrt{s^2}} = \frac{\bar{Y} - \mu_0}{s/\sqrt{n}}$$

The one-sample  $t$  test is a LR test!!!